UNIVERSITI MALAYSIA SARAWAK

Faculty of Computer Science and Information Technology

TMA3084 Software Engineering Laboratory | Semester 1, 2026/2027

PROJECT PROPOSAL

(Product Requirements Document)

<Event & Attendance Tracking App>

Group Number / Name	Group 21 – ByteChew
Lab Group / Demonstrator	Group 2 – Mr. Raynold anak Kabai
Client / Organisation	Lambert Gan, Chief Executive Officer, Skybylamby
Lecturer	Mdm. Nurfauza Jali
GitHub Repository	https://github.com/afqskb/TMA3084_G02_T21_ByteChew_EventnAttendanceTrackingApp
Version / Date	v1.0 (original) – 17/10/2026

Team Members

No.	Name	Matric No.	TSPi Role	Signature
1	Muhammad Afiq bin Sakub	102719	Team Leader, Planning Manager	
2	Adenillo Sheldy anak Stephen	102372	Support Manager	
3	Ahmad Danish bin Alin	102384	Quality Manager	
4	Shelsy anak Nicholas Felix	102958	Development Manager	
5	Syafik bin Sali	102978	Client Liaison	

Declaration

By signing above, we certify that this proposal is our own work, that all external sources, libraries, APIs and AI-assisted content are acknowledged, and that we have not engaged in plagiarism or unauthorised collusion (Peraturan Akademik UNIMAS, Etika Akademik). AI tool usage is recorded in LOGAI.

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Abstract

PRD section: Executive Summary / TL;DR **Rubric link:** Clarity & presentation

Universities host many academic, cultural, and social activities, but events are mostly promoted through scattered social media posts, so students struggle to discover relevant events and forget to attend, while organisers can only guess how many people will come. ByteChew addresses this with a centralised Android app built with Flutter and Firebase. It offers an interactive campus map with event locations, one tap attendance confirmation with a live attendee count for organisers, a push reminder a week before the event, and AI filtering and sorting that ranks events by each students' interests. The project is complete when a signed release APK (v1.0.0) is delivered, all core feature tests pass, a SUS of 70 or above is achieved with at least 20 testers, and the client signs off in User Acceptance Testing (UAT).

1. Introduction

PRD section: Background & Context **Rubric link:** Background (Section A)

Universities host academic, recreational, cultural and social events, and information about them is usually shared across many social media and messaging channels. This makes it hard for students to find and keep track of relevant events, and they often miss events that have already been announced. Organisers face the opposite problems, since they cannot reliably tell how many people will attend. A centralised event discovery, location information, attendance registration and notifications in one place. It can also let authorised students create and manage their own events, encouraging participation across the university community.

1.1 Client Background

Skybylamby, founded in 2025, is a dynamic creative media production service provider led by Chief Executive Officer Lambert. The organisation specialises in delivering high-quality media production solutions, catering to a growing audience and client base expected to scale beyond 10,000 users.

1.2 Team Background

ByteChew is a team of five 3rd year students whose motto, "Chewing on problems, one byte at a time," reflects on how we approach the project by breaking a large problem into smaller manageable steps. Our skills cover Flutter development, Firebase, AI API integration, testing

and project management, and each member takes a TSPi role (Humphrey, 2000) so that planning, development, quality, support and client communication are all covered.

2. Problem Statement

PRD section: Problem & Opportunity **Rubric link:** Objectives / constraints analysed

ID	Current process / problem	Impact on client	Evidence (source)
P1	Organisers promote events by posting on social media, and students must scroll through many posts to find relevant events	Low awareness and engagement	Client interview, 10/10 (ByteChew, 2026)
P2	Organisers rely on guesses or informal responses to estimate attendance	Wasted resources or overcrowding	Client interview, 10/10
P3	Event information is scattered across different posts and platforms	Attendees are confused and miss events	Client interview, 10/10
P4	Students receive no reminders as the event date approaches	Lower turnout	Client interview, 10/10

Table 1: Summary of client problems

A Flutter mobile app that maps university events with attendance booking and timely reminders, using AI to recommend events to students based on their interest.

3. Project Objectives & Success Metrics

PRD section: Goals & Success Metrics **Rubric link:** Objectives well defined and prioritised

No.	Objective	Solves	Success metric (how we measure)	Priority
O1	To gather requirements by interviewing the client about how they find, promote and plan for events	P1, P2, P3, P4	At least 1 client interview completed	High

No.	Objective	Solves	Success metric (how we measure)	Priority
O2	To build the Flutter app with an interactive campus map, event details and the “Confirm Attendance” button	P1, P2, P3	All event locations on the map show details; a booking updates the organiser’s attendee count in real time with 100% accuracy in test cases	High
O3	To add notifications 1 week before an event, and an AI powered filtering and sorting of events by student interests	P4	Reminders delivered for ≥ 95% of booked events; relevant events in the top results for ≥ 80% of AI test queries	Medium
O4	To conduct user testing with students and organisers and evaluate whether the app solves the problems	P1, P2, P3, P4	≥ 20 testers; SUS ≥ 70 (Brooke, 1996); ≥ 80% of students find events easier to find; ≥ 80% of organisers say counts help planning; client sign-off on the final demo	High

Table 2: Project objectives and success metrics

4. Project Perspective

PRD section: Target Users & System Context **Rubric link:** Information obtained and analysed

4.1 Target Users

User type	Description	Main needs / goals	Device / tech level
Student	University students	Find events by AI sorted list or map, get reminders, and confirm attendance	Android phone, basic

User type	Description	Main needs / goals	Device / tech level
General Public	Visitors, alumni, or parents	Find public events and navigate the campus	Android phone, basic
Event Organiser	Club committee members or staff	Post events and see confirmed attendee counts	Android phone, intermediate
System Admin	University staff managing the app	Approve, reject or remove events, manage organiser accounts	Android phone, intermediate

Table 3: Target users

4.2 System Context and Architecture

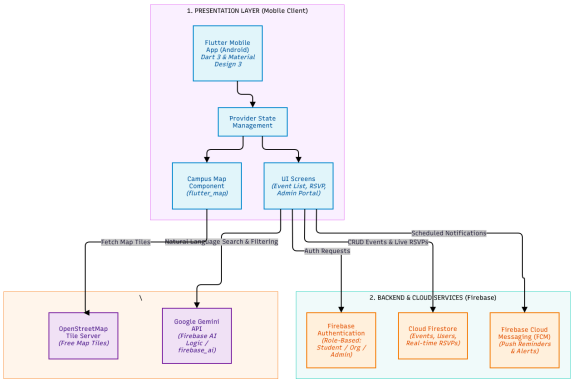


Figure 1: High-level system architecture

As illustrated in Figure 1, the tracking app operates on multi-tier architecture. The client-side Presentation Layer is built with Flutter and Dart 3 using Material Design 3 (Google, n.d-b), and Provider (Rousset, n.d.) state management featuring core UI screens and a flutter_map (fleaflet, n.d.) campus map component which fetches tiles from the OpenStreetMap Tile Server (OpenStreetMap Foundation, n.d.). Backend operations integrate with Firebase, utilising Firebase Authentication for role-based access control (Student, Organiser, Admin), Cloud Firestore for real-time event data and RSVP persistence, and Firebase Cloud Messaging for automated push notifications (Google, n.d.-a). Additionally, intelligent event discovery and natural-language filtering via Google Gemini API (Google, n.d.-c) accessed via Firebase AI Logic (Google, n.d.-d) and the firebase_ai package (Firebase, n.d.-a).

5. Project Scope

PRD section: Scope, Features & Requirements **Rubric link:** Design recommendations supported

5.1 In Scope / Out of Scope

In scope (will deliver this semester)	Out of scope (will not deliver)
Android mobile app built with Flutter	iOS version and publishing to the Google Play Store
Basic admin interface to approve, reject or remove events	Full admin web portal with analytics and reports
Interactive campus map with event markers	Indoor navigation and turn-by-turn directions
Organiser can create an event and place them on the map	Event analytics dashboards and feedback forms
“Confirm Attendance” button with live attendee count	QR code check-in, waiting lists and seat selection
Push notification reminder 1 week before a booked event	SMS or email reminders and multiple reminder times
AI sorted event list as the home screen, filtered by student interests	Training our own AI model or predicting turnout
Firebase login with roles	University single sign-on integration
Firebase backend for data, authentication and notifications	Custom server and data migration
Testing with at least 20 users, a SUS survey and a client demo	Large-scale load testing and security audits

Table 4: Scope boundary

5.2 Features and User Stories

FR-ID	Module	User story (As a … I want … so that …)	Priority
FR-01	User Management	As a student, I want to log in with my email so that I can book events and get reminders.	Must
FR-02	User Management	As a system, I want to assign each account a role so that users only see features they are allowed to use.	Must
FR-03	User Management	As a user, I want to reset my password so that I can regain access if I forget it.	Could
FR-04	Campus Map	As a student, I want to see a campus map with event markers so that I know where events are held.	Must
FR-05	Campus Map	As a student, I want to tap a marker to see the event details so that I can decide quickly if it interests me.	Must
FR-06	Event Discovery	As a student, I want to see a list of upcoming events so that I can browse without using the map.	Should
FR-07	Event Discovery	As a student, I want to filter events by date or category so that I can find relevant events faster.	Could
FR-08	Event Discover	As a general public visitor, I want to log in with a non-student email so that I can browse events open to the public.	Should
FR-09	Event Management	As an organiser, I want to create an event with its details and map location so that students can discover it in one place.	Must
FR-10	Event Management	As an organiser, I want to edit or cancel my event so that students always see correct information.	Must
FR-11	Event Management	As an organiser, I want to upload a poster so that my event looks attractive.	Could

FR-ID	Module	User story (As a … I want … so that …)	Priority
FR-12	Event Management	As an event organiser, I want to mark my event as public or students only so that the right audience sees it.	Should
FR-13	Attendance Confirmation	As a student, I want to confirm my attendance with one tap so that organisers know I am coming.	Must
FR-14	Attendance Confirmation	As a student, I want to cancel my attendance so that the attendee count stays accurate.	Must
FR-15	Attendance Confirmation	As a student, I want to see my booked events so that I can keep track of my plans.	Should
FR-16	Attendance Confirmation	As an organiser, I want to set a maximum capacity so that bookings stop when the venue is full.	Could
FR-17	Notifications	As a student, I want a push notification 1 week before a booked event so that I do not forget to attend.	Must
FR-18	Notifications	As a student, I want a notification when a booked event changes or is cancelled so that I do not go to the wrong place or time.	Could
FR-19	Admin Management	As an admin, I want to approve or reject submitted events so that only appropriate events appear.	Must
FR-20	Admin Management	As an admin, I want to manage organiser accounts so that only authorised users can create events.	Should
FR-AI-01	AI Feature	As a student, I want to set my interests so that the AI sorts the most relevant events first.	Must

Table 5: Features and user stories

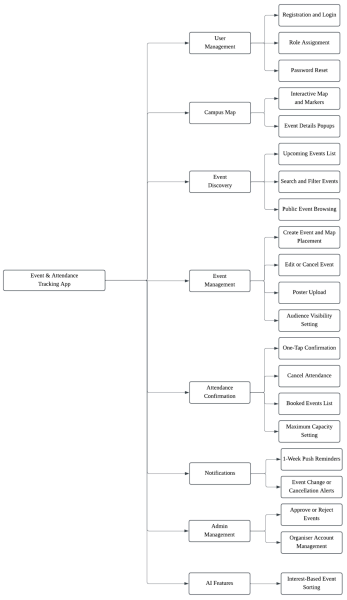


Figure 2: System modules diagram

5.3 AI Feature

This app uses AI to filter and sort upcoming events by a student's saved interests or a typed phrase, ranking the most relevant first. This solves the problem of students scrolling through many posts to find the few events that matter to them. The AI only sorts events already in the database, so it cannot invent events.

Item	Description
AI capability	Filtering and sorting upcoming events by relevance from interests or natural-language requests
User value	Solves P1 which is students find relevant events faster, and organisers reach the students most likely to attend
Input → Output	Input: the student's interests or typed request. Output: a list ordered from most to least relevant
AI service / API	Gemini API (Google, n.d.-c) via Firebase AI Logic and firebase_ai (Firebase, n.d.-a)
Constraints	API key stays out of the app. If free tier limits are reached, the app falls back to date and category filter. Only event data is sent to the AI, with no personal student data.

Table 6: AI feature summary

5.4 Non-Functional Requirements and Constraints

NFR-ID	Category	Requirement / constraint	Target
NFR-01	Performance	The main screens and AI filtering respond quickly.	Map and event list load in < 3 s on a mid-range Android phone over 4G; AI filtering responds in < 5 s for ≥ 90% of test queries.
NFR-02	Security	API keys are never stored in the app or source code, each role can only use its own features, and only the minimum personal data is sent to the AI service.	0 secrets repository or app package; 100% of role tests pass; only interests, typed requests and event data sent to the AI, with no names or emails.
NFR-03	Usability	A first-time student can use the core features without training.	SUS score of ≥ 70 (Brooke, 1996); a new user can open an event and confirm attendance within 30 s and 3 taps with no instructions in ≥80% of attempts.
NFR-04	Reliability	Attendance data stays correct, reminders are delivered, and the app keeps working if the AI service is down.	Attendee count is 100% accurate in test cases; ≥ 95% of reminders delivered; on AI failure the normal event list and search are shown instead of an error.

Table 7: Non-functional requirements and constraints

5.5 System Environment

Layer	Technology / tool	Justification
Mobile app	Flutter (Dart 3), Material 3	Course requirement, and one codebase for Android
State management	Provider	Simple and enough for this app size

Backend / database	Firebase Authentication and Cloud Firestore (Firebase, n.d.-a, n.d.-c, n.d.-d)	Email login, roles and real-time attendee counts without our own server
Notifications	Firebase Cloud Messaging and flutter_local_notifications (Firebase, n.d.-e; Flutter Community, n.d.)	Sends 1 week reminder and change or cancel alerts
Map	flutter_map with OpenStreetMap	Free campus map with no billing account
AI service	Gemini API via Firebase AI Logic, firebase_ai package	Keeps the API key out of the app and free tier for testing
Version control & CI	GitHub, GitHub Actions (GitHub, n.d.-a)	Course requirement
Target devices	Android 8.0+ phones	Covers common Android phones since iOS is out of scope

Table 8: System environment

6. Task Allocation & Team Contribution

PRD section: Team & Stakeholders (RACI) **Rubric link:** Project breakdown; team work

Member	TSPi role	Owned features (FR-IDs)	Other responsibilities / contribution
Muhammad Afiq bin Sakub	Team Leader, Planning Manager	FR-01, FR-02, FR-03, FR-20	Weekly meetings, WEEK reports, TASK / SCHEDULE, GitHub board
Shelsy anak Nicholas Felix	Development Manager	FR-04, FR-05, FR-06, FR-07	Architecture, code review
Ahmad Danish bin Alin	Quality Manager	FR-13, FR-14, FR-15, FR-16	LOGTEST, CI, testing

Member	TSPi role	Owned features (FR-IDs)	Other responsibilities / contribution
Adenillo Sheldy anak Stephen	Support Manager	FR-17, FR-18, FR-08, FR-AI-01	Repo setup, LOGAI, tools
Syafik bin Sali	Client Liaison	FR-09, FR-10, FR-11, FR-12, FR-19	Client meetings, UAT

Table 9: Task allocation and contribution

7. Team Goals

PRD section: Team Working Agreement **Rubric link:** Team work

No.	Team goal	Measure
G1	Deliver a quality product that solves the client's problems	All Must-have FRs pass their tests; CI green on main; SUS score of 70 or above
G2	Run a productive, well-managed project	Every milestone delivered on or before its date; GitHub board updated before each weekly meeting
G3	Make sure every member contributes fairly	Each Member merges at least 1 PR per sprint and completes their owned FRs
G4	Keep the client engaged	Client meeting at least every 2 weeks; client approves requirements and signs off the final demo

Table 10: Team goals

8. Expected Results

PRD section: Release Criteria / Deliverables **Rubric link:** Decisions supported by information

8.1 Working System

The final system is an Android app, built with Flutter and backed by Firebase, that gives the students and the general public one place to discover university events. The home screen is a list of upcoming events, filtered and sorted by AI so that a student's saved interests, or a phrase they type, bring the most relevant events to the top. Tapping an event shows its details

and a “Confirm Attendance” button, and students receive a push reminder one week before events they have booked. A second tab shows the interactive campus map, with a marker at each event location. General public visitors can only confirm attendance events marked as open to the public. Organisers can create, edit and cancel events, set them as public or students-only, and see a live count of confirmed attendees. The system admin can approve, reject or remove events and manage organiser accounts. This system is ready for release when a signed release APK (v1.0.0) is produced, all Must-have features pass their tests on the main branch, user testing reaches a SUS score of 70 or above with at least 20 testers, and the client signs off the final demo in UAT.

8.2 Documentation and Artefacts

Deliverable	Submitted with
Project Proposal (original)	Proposal Submission (Week 4)
Revised Proposal and Software Requirements Specification (SRS)	Initial Report (Week 9)
Runnable prototype (4–5 screens), tagged release	Demo 1 (Week 9)
Software Design Specification (SDS), test plan and results (LOGTEST), user manual, maintenance plan, client acceptance form	Final Report (Week 13)
Release APK, demo video (5–10 min), presentation slides	Final Demo 2 (Week 14)
GitHub repository and complete project file	e-Archive (Week 15)
TSPi forms (LOGT, LOGD, LOGAI, SCHEDULE, SUMP, SUMQ, SUMS, SUMTASK, WEEK, PEER) and meeting minutes	Ongoing; compiled in Final Report

Table 11: Deliverables

9. Project Planning

PRD section: Milestones, Timeline & Risks **Rubric link:** Methodology to be followed

9.1 Methodology

The project follows an Agile development approach structured around 2-week iterative sprints, aligning with the dynamic nature of client requirements and the 15-week academic timeline. Development tasks are managed transparently using GitHub Projects boards (GitHub, n.d.-a) and Issues assigned to team members according to their TSPi roles. Every feature implementation follows a strict spec-driven workflow: requirements are mapped from the Product Requirements Document (PRD), developed within isolated feature branches, subjected to mandatory peer code reviews via Pull Requests (PRs), and verified through automated CI checks on GitHub Actions (GitHub, n.d.) before merging into the main branch.

9.2 Submission Schedule

Phase	Deliverable	Week	Due
Phase 1: Planning & Development	Proposal Submission	4	16 Oct 2026
	Proposal Presentation	5	19 – 23 Oct 2026
	Initial Report	9	16 Nov 2026
	Demo 1	9	16 – 20 Nov 2026
Phase 2: Finalisation & Closure	Final Report	13	24 Dec 2026
	Final Demo 2	14	28 – 31 Dec 2026
	e-Archive	15	Before 10 Jan 2027

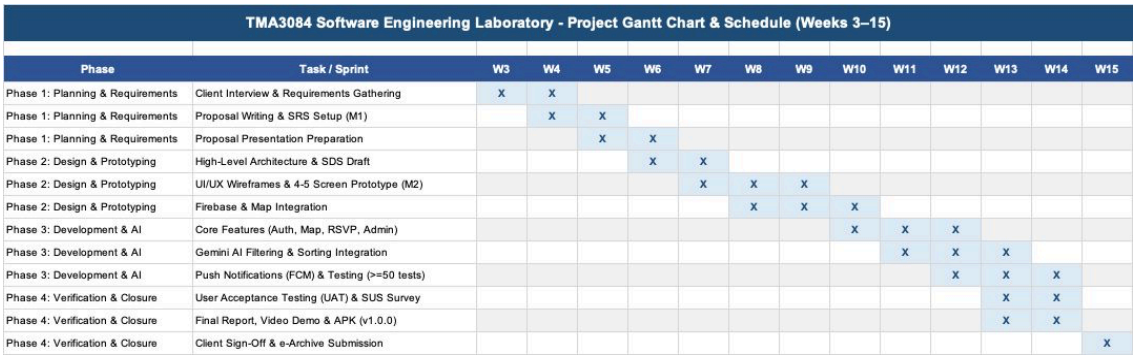
Table 12: Submission schedule

9.3 Development Milestones (GitHub)

Milestone	Week	Key output	Status
M1 – Scope & Requirements	4 – 5	GitHub repo, issue board, CI file; proposal and initial SRS in /docs	In Progress

Milestone	Week	Key output	Status
M2 – Design & Prototype	9	SDS draft, 4–5 screen prototype, green CI, v0.2.0 tag	Pending
M3 – Product & Verification	13	≥50 tests, ≥60% coverage, AI feature working	Pending
M4 – Final Release & Acceptance	15	Release APK v1.0.0, client sign-off, e-Archive	Pending

Table 13: Development milestones



TMA3084 Software Engineering Laboratory - Project Gantt Chart & Schedule (Weeks 3–15)

Phase	Task / Sprint	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14	W15
Phase 1: Planning & Requirements	Client Interview & Requirements Gathering	X	X											
Phase 1: Planning & Requirements	Proposal Writing & SRS Setup (M1)		X	X										
Phase 1: Planning & Requirements	Proposal Presentation Preparation			X	X									
Phase 2: Design & Prototyping	High-Level Architecture & SDS Draft				X	X								
Phase 2: Design & Prototyping	UI/UX Wireframes & 4-5 Screen Prototype (M2)					X	X	X						
Phase 2: Design & Prototyping	Firebase & Map Integration						X	X	X					
Phase 3: Development & AI	Core Features (Auth, Map, RSVP, Admin)								X	X	X			
Phase 3: Development & AI	Gemini AI Filtering & Sorting Integration									X	X	X		
Phase 3: Development & AI	Push Notifications (FCM) & Testing (>=50 tests)										X	X	X	
Phase 4: Verification & Closure	User Acceptance Testing (UAT) & SUS Survey											X	X	
Phase 4: Verification & Closure	Final Report, Video Demo & APK (v1.0.0)											X	X	
Phase 4: Verification & Closure	Client Sign-Off & e-Archive Submission													X

Figure 3: Gantt chart

9.4 Risks and Assumptions

ID	Risk / assumption	Likelihood	Impact	Mitigation
R1	Client delays feedback or UAT sign-off	Medium	High	Fixed fortnightly meeting with Lambert Gan (Skybylamby) plus an online channel
R2	Gemini free tier limits reached during testing	Medium	High	Cache queries and fall back to date and category filters
R3	Skill gaps with Provider, flutter_map or Firebase	Medium	High	Pair programming and peer code reviews
R4	Scope creep delays delivery	Low	Medium	Follow the scope and build Must-have stories first

Table 14: Risks and assumptions

10. References

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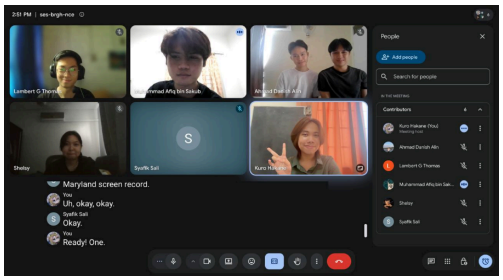
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Appendix A: Client Meeting Evidence

Item	Details
Date	10 October 2026
Attendees	1. Muhammad Afiq bin Sakub 2. Ahmad Danish bin Alin 3. Adenillo Sheldy anak Stephen 4. Shelsy anak Nicholas Felix 5. Syafik bin sali 6. Lambert (Client)
Key Decisions	1. Build a centralised mobile app that bridges the information gap between organisers and attendees and gives organisers a live count of confirmed attendees. 2. Target users include attendees (students and general public), organisers, and system admin for approving events and managing accounts. 3. Core features in scope include 1 week push notification reminder, interactive campus map with event markers and a confirm attendance button with a live attendee count. 4. Multiple reminder times and QR check-in and analytics dashboards are out of scope. 5. Client requests blue as the primary colour of the application and black as the secondary colour. Client also note that AI chatbots are highly discouraged, so AI features are limited to interest-based filtering and sorting of events only.
Photo Evidence	

Appendix B: Revision History

Version	Date	Changes (e.g. after proposal presentation feedback)	By
1.0	17/10/2026	Original submission	Danish

Appendix C: Recorded Video Pitch

Video link: https://youtu.be/tNUCElRzwpg